

BUSINESS CASE

Document Information	
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Purpose:

This Business Case provides the justification for implementing the Student Performance Analytics & Decision Support System at Mega Central High School. It outlines the business need, current challenges, proposed solution, expected benefits, investment considerations, project risks, and recommendations to support informed decision-making by school management.

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1. Executive Summary:

Mega Central High School currently manages Grade 10 assessment data using separate spreadsheets maintained by individual subject teachers. While this approach supports basic record keeping, it limits the school's ability to monitor academic performance across subjects, identify students requiring timely intervention, evaluate chapter-level learning outcomes, and generate consolidated reports efficiently.

To address these challenges, the school proposes implementing a Student Performance Analytics & Decision Support System. The solution will centralize academic data, automate score calculation, track student progress over time, identify at-risk students, and provide interactive dashboards for teachers and school management.

The proposed system is expected to significantly reduce manual reporting effort, improve reporting accuracy, support data-driven academic planning, and enable proactive interventions that enhance student learning outcomes.

2. Business Need:

The current academic reporting process depends heavily on manually maintained spreadsheets across multiple subjects. This decentralized approach results in inconsistent reporting practices, duplicated effort, delayed academic insights, and limited visibility into overall student performance.

School management requires a centralized analytics platform capable of consolidating assessment data, generating standardized reports, monitoring student progress continuously, and providing actionable insights that support evidence-based academic decision-making.

3. Problem Statement:

Mega Central High School lacks a centralized academic performance monitoring system. Student assessment data is distributed across multiple Excel files maintained independently by subject teachers, making report preparation time-consuming and prone to inconsistencies.

As a result, academically weak students are often identified only after final examinations, chapter-level learning gaps remain unnoticed until late in the academic cycle, and school leadership has limited access to real-time academic performance information required for timely intervention and strategic planning.

4. Current Challenges:

Current Challenge	Business Impact
Student data stored in multiple Excel files	Data inconsistency and duplication
Manual score calculation	Increased teacher workload
Manual report preparation	Delayed academic reporting
No centralized dashboard	Limited visibility for school management
Weak students identified late	Delayed academic intervention
No chapter-wise analysis	Difficulty identifying learning gaps
No historical trend analysis	Limited ability to monitor student improvement

5. Root Cause Analysis:

Primary Causes:

People

- Teachers maintain separate spreadsheets.
- Reporting methods differ across departments.
- Manual consolidation introduces errors.

Process

- No standardized reporting process.

- Manual score aggregation.
- Reports generated only after examinations.

Technology

- Reliance on Excel spreadsheets.
- No centralized database.
- No analytical dashboards.
- No automated calculations.

Data

- Duplicate student records.
- Inconsistent file formats.
- Missing validation checks.
- Limited historical performance tracking.

These factors collectively reduce reporting efficiency and delay academic decision-making.

6. Proposed Solution:

Develop a centralized Student Performance Analytics & Decision Support System that will:

- Consolidate student assessment data into a unified repository.
- Automate score calculation and grading.
- Track student performance across multiple assessments.
- Identify students at academic risk.
- Analyze subject-wise and chapter-wise performance.
- Provide executive dashboards through Microsoft Power BI.
- Support teachers with actionable academic insights.
- Enable management to monitor institutional performance in real time.

7. Business Benefits:

Tangible Benefits

Benefit	Expected Outcome
Automated score calculation	Eliminate manual calculations
Faster report generation	Reduce reporting effort by approximately 80%
Centralized reporting	Single source of academic data
Real-time dashboards	Immediate access to performance insights
Standardized reporting	Improved reporting consistency

Intangible Benefits

- Improved academic decision-making.
- Earlier identification of at-risk students.
- Better collaboration among teachers.
- Increased confidence in academic reports.
- Improved curriculum planning.
- Enhanced teaching effectiveness through chapter-level analytics.

8. Cost–Benefit Analysis:

Estimated Investment

Category	Assessment
Software	Low (Open-source tools and Power BI Desktop)
Development Effort	Moderate
Training	Low
Maintenance	Low

Expected Benefits

Benefit	Business Value
Reduced teacher workload	High
Faster decision-making	High
Improved reporting accuracy	High
Early student intervention	High
Better academic planning	High

Overall, the anticipated benefits significantly outweigh the implementation effort, making the project a high-value investment for the school.

9. Alternatives Considered:

Option	Advantages	Disadvantages
Continue using Excel	No implementation effort	Manual, inconsistent, limited analytics
Purchase commercial school analytics software	Rich functionality	High licensing and customization cost
Develop custom analytics solution	Tailored to school requirements, scalable, cost-effective	Initial development effort required

After evaluating the available options, the custom Student Performance Analytics & Decision Support System is recommended as it best aligns with the school's operational requirements, budget considerations, and long-term academic goals.

10. Risks:

Risk	Impact	Mitigation
Incorrect assessment data	High	Implement validation rules and periodic data quality checks
Missing student responses	Medium	Generate data quality reports and exception logs
User adoption challenges	Medium	Conduct teacher training and provide user guides
Scope changes	Medium	Follow formal change management procedures
Data privacy concerns	High	Implement role-based access and secure data storage

11. Recommendation:

Based on the identified business challenges, expected operational improvements, and favorable cost–benefit assessment, it is recommended that Mega Central High School proceed with the implementation of the Student Performance Analytics & Decision Support System.

The proposed solution will centralize academic reporting, automate manual processes, improve visibility into student performance, support early identification of academically at-risk students, and enable data-driven decision-making across the institution. Successful implementation is expected to improve reporting efficiency, enhance teaching effectiveness, and contribute to better student learning outcomes.

12. Approval:

Role	Name	Signature	Date
Chairman			
Principal			
Project Manager	Phanikrishna		
Business Analyst	Ramyashree GV		